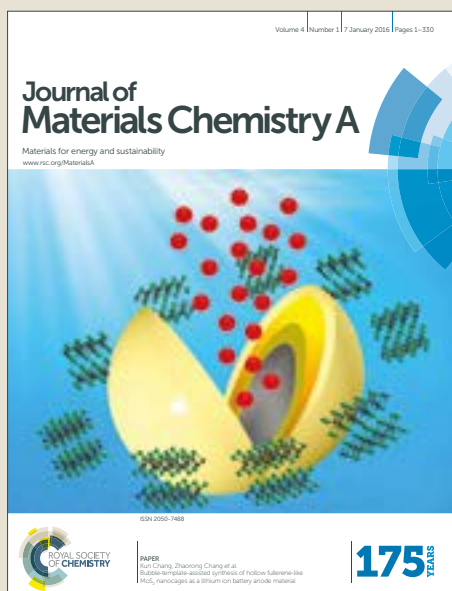


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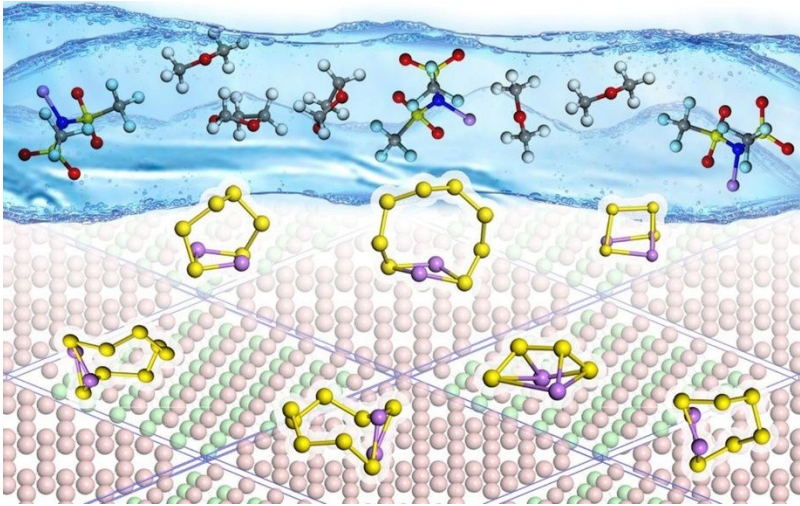
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Table of Contents



The polarized surfaces of borophenes provide appropriate sites to immobilize polysulfides, even under 300 K and high sulfur loading.



Journal Name

ARTICLE

Interfacial Competition between Borophene-Based Cathode and Electrolyte for the Multiple-Sulfide Immobilization of Lithium Sulfur Battery†

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The practical application of lithium-sulfur battery (LSB) is mainly hindered by the poor conductivity of the sulfur cathode, polysulfide shuttle, and volumetric variation during cycling. To this end, host materials with high conductivity and excellent polysulfide absorbability extensively employed as the loading scaffolds for sulfur in LSB. In the present work, three types of borophene nanosheets with metallicity are theoretically evaluated as the host for sulfur cathodes. The polarization of surface, conductivity, polysulfide immobilization, and the interfacial competition between borophenes and electrolytes are particularly discussed. The strong binding of polysulfides on borophenes will block the shuttle effect of polysulfides, due to the strong Coulomb interactions between B and S atoms, even considering the competition effect of the electrolytes and under high sulfur loading. The calculated results clearly indicate that borophene-based materials with the polarized and well conductive surfaces hold great promise for LBS.

Introduction

To address the intrinsic drawbacks such as poor conductivity, the shuttle effect induced by polysulfide (Li_2S_n , $n=4,6,8$) dissolution, and great volume fluctuation of sulfur cathode in charge/discharge processes and improve the performance of lithium-sulfur battery (LSB), many advanced strategies have been developed,¹⁻³ by adding conductive additives, polysulfide absorbents, and/or the design of porous configurations. Even so, the LSB performance so far still cannot reach the requirements for practical applications. Many works demonstrated^{1-2, 4-5} the nonpolar feature of sp^2 carbon cannot satisfy the full immobilization of polysulfide during cycling, resulting in the loss of sulfur. And poor conductivity of polar materials including metal oxides and metal sulfides hindered their applications.⁶⁻¹⁰ Therefore, searching new materials with excellent conductivity and superior absorbability are still highly desirable.

Among the previously studied materials, two-dimensional (2D) host materials are regarded as one of the most promising

alternatives for LSBs,¹¹⁻²⁹ especially conductive host materials. For instance, graphene has been used as a conductive host material in LSBs and exhibited improved performance.¹¹⁻¹⁴ Additionally, introducing heteroatoms/functional-groups in graphene sheets¹⁵⁻¹⁸ or other carbon materials could partially change the non-polar behaviour.¹⁹⁻²² Still, the weak interaction between S and C atoms cannot efficiently adsorb Li_2S_n on cathode.^{1, 4-5} In addition to graphene, layered MXene with high conductivity and strong binding strength with polysulfides are also explored as a potential 2D conductive host.²³⁻²⁵ However, pure MXene nanosheets are difficult to exfoliate, and many functional groups on their surfaces cannot be removed easily.²⁶ Besides, too strong interaction from bare MXenes will result in dissociation of Li_2S_n clusters,²⁴ and in fact the heavy transition metal atoms are also not ideal to develop LSB with high gravimetric energy density. It should be noticed that, in theoretically, the adsorption of single Li_2S_n was frequently used to evaluate the performance of host materials. Actually, the practical application system of LSB must have high sulfur loading, which of them should be dissolved by electrolyte. Therefore, for practical applications, it is essential to explore the properties of high sulfur loading as well as their mechanism. More than that, the influence of electrolytes under practical conditions is also imperative, especially why the electrolytes can or not dissolve the polysulfides.

Recently, a series of new 2D conductive materials, known as borophene family, including triangular borophene (ΔB),³⁰ β_{12} -borophene ($\beta_{12}\text{B}$),³¹ χ_3 -borophene ($\chi_3\text{B}$)³¹ have been successfully synthesized. Both experimental and theoretical investigations have demonstrated that these borophenes possess metallicities,³⁰⁻³⁴ and also have structural channels for fast ion

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† Electronic Supplementary Information (ESI) available: Electronic structures of borophenes and sulfides adsorbed ones, the geometric structures of one or more Li_2S_n on borophenes, optimized structures of Li_2S_n around solvents and binding energies of DOL and DME on borophenes, and a test of the mass loading of sulfur on graphene. See DOI: 10.1039/x0xx00000x

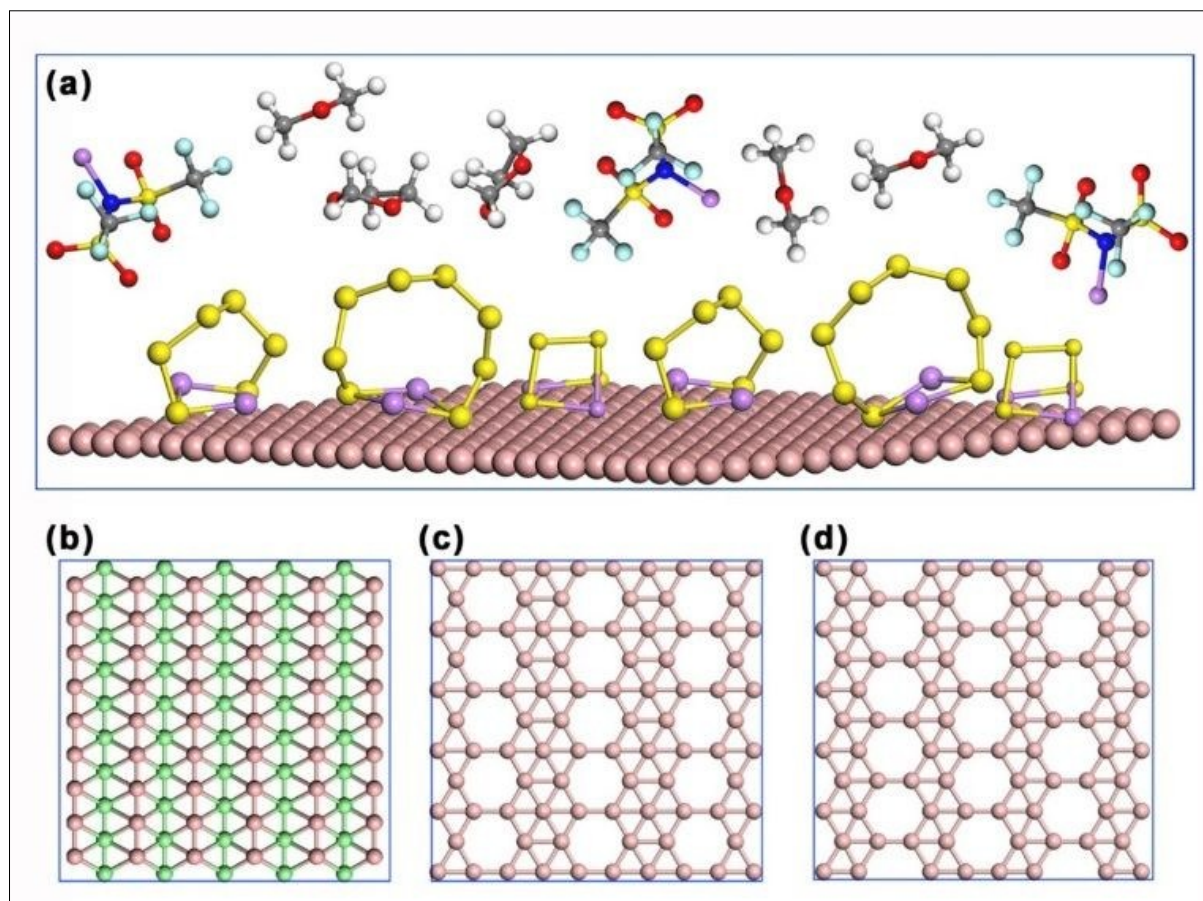


Fig. 1 (a) The entrapment of polysulfides by borophene against electrolyte and the structural models of (b) triangular borophene (ΔB), (c) β_{12} -borophene ($\beta_{12}B$), (d) χ_3 -borophene (χ_3B). Pink balls, B; yellow balls, S; purple balls, Li; gray balls, C; red balls, O; light-blue balls, F; white balls, H; dark balls, N. For ΔB , the pink and green balls respectively stand for the groove and bulge B atoms.

migration.^{32–34} Similar to graphene, they also own high specific areas for sulfur loading. Since the presence of porous or wave-type configuration, these borophene sheets have polar structures that are ideal to confine Li_2S_n .^{6, 35–37} Moreover, boron atoms are very light, making it beneficial for high energy densities as well as power densities of batteries. Recently, few theoretical studies of borophenes for LSB are reported,^{38–40} in which single Li_2S_n adsorption and its properties are explored. However, the practical conditions have not been considered in these works, including high sulfur adsorption and the influence of electrolytes.

Herein, we theoretically explore the potential application of the borophene sheets as the host materials for sulfur cathodes (Fig. 1) based on first principle calculations. Focused on the interactions among borophene sheets, lithium sulfides and

electrolytes, we systematically discuss the influence of the polarized surface and conductivity of borophenes on sulfides immobilizations, especially the interfacial competition between borophenes and electrolytes. The borophenes as the polysulfide absorbents could potentially block the dissolution of sulfur cathode by the strong Coulomb interactions between B and S atoms induced by the charge redistribution. Besides, the high sulfur loading on borophenes are also minutely discussed, ~80% sulfur can be effectively loaded on borophenes.

Results and discussion

We first investigated the electronic structures of three kinds of borophene sheets, including the wave-like configuration of ΔB ,

the hexagonal porosity of $\beta_{12}\text{B}$, and the $\chi_3\text{B}$. The charge density distributions of them are shown in Fig. 2, where the obviously unequivalent charge density indicates that the existence of electronically polarized sites. Specifically, electrons are more likely located at the crest of ΔB , or around pore edges of $\beta_{12}\text{B}$ and $\chi_3\text{B}$, which means that Li_2S_n clusters could be easily anchored on these sites. The polarized surface of host materials can assist cathodes to confine polysulfides, and thus achieve better LSB performance. Moreover, the borophene sheets were also confirmed to have good metallicity (Fig. S1†)^{6, 35–37}, which is beneficial for both anodes and cathodes in lithium ion batteries.

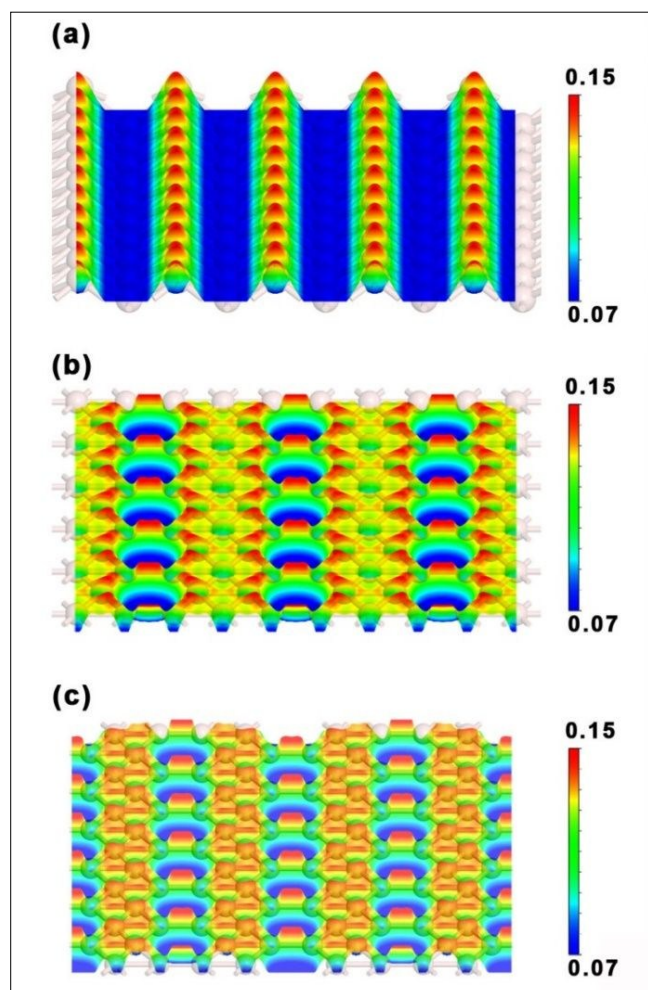


Fig. 2 Charge densities (in $\text{e}/\text{\AA}^3$) of (a) ΔB , (b) $\beta_{12}\text{B}$ and (c) $\chi_3\text{B}$ surfaces.

The shuttle effect induced by dissolution of polysulfides can heavily decay the practical performance of LSBs. The loss of sulfur will deactivate the cathode with increasing the density of S_n^{2-} in electrolyte, some of which across the separator and further react with Li anode driven by the electric field between cathode and anode. As a result, a lot of non-conductive by-products are generated and covered on anode, which will impede the Li ion diffusion and reduce the electrochemical kinetics.^{41–42} One of the reasons for the shuttle effect is caused by the weak binding with host materials and strong attraction

by electrolytes. The interaction between sulfides and borophene sheets as well as the attraction by electrolytes were thus investigated. The 1,3-dioxolane (DOL) and 1,2-dimethoxyethane (DME) were adopted as the electrolyte solvents. As shown in Fig. 3a, the binding energies of Li_2S_n on all borophene sheets are larger than those on DOL or DME, suggesting the borophenes surface can effectively adsorb the Li_2S_n . Specifically, the adsorption energy values of Li_2S_n on borophenes are larger than 2.0 eV, except Li_2S_4 and Li_2S_6 on $\chi_3\text{B}$ (1.86 eV and 1.28 eV respectively), while the corresponding binding energies of Li_2S_n on DOL or DME are less than 1.0 eV. For long-chain Li_2S_n ($n = 4, 6, 8$), the strong binding on borophenes can prevent them from dissolving, thus reduce the loss of sulfur and alleviate the shuttle effect. For end-products (Li_2S and Li_2S_2), the strong binding with host materials can inhibit their precipitation in solvent by minimizing the loss of sulfur and further maintaining the charge/discharge specific densities. On the other hand, the distances between S and borophenes are no longer than 2 Å (Fig. S2† and Fig. S3†), which are smaller than those between S and DOL or DME, also suggesting the strong interactions between S and B atoms. And what's more, the nearest B atoms to sulfurs are mainly at the crests of ΔB , or around the pores of $\beta_{12}\text{B}$ and $\chi_3\text{B}$, which further demonstrates that the polar surfaces are beneficial to entrap Li_2S_n . Besides, all long-chain Li_2S_n ($n=4, 6, 8$) are distorted, which is also attributed to the strong adsorption between the polysulfides and borophenes. Fig. 3a also shows the desirable binding energies of pure sulfur (S_8) with three kinds of borophenes sheets are all more than 0.5 eV, which suggests the initial active materials can be maintained on the surface. Interestingly, it is observed that the surfaces of borophene sheets will not be occupied by DOL or DME, which is attributed to that sulfides being first adsorbed on the host materials in process of battery assembling with stronger binding energies than that on DOL or DME (~ 0.3 eV, as listed in Table S1†).

For all kinds of energy storage systems, the gravimetric or volumetric energy densities should be one of the most important parameters for their practical applications. To maintain the capacity advantage of LSBs, the mass of sulfur should be higher than 70 wt% in original cathodes.⁴³ Therefore, to satisfy the requirement in practical use, the contents of sulfur are in the range of 60~80 wt% in recent literatures,^{44–46} which means that there are plenty of sulfides on surfaces of host materials during cycling. The question is whether the host can hold enough sulfides, which has not been deeply studied elsewhere. In this work, in order to characterize the potential application of borophenes in LSBs, multiple Li_2S_n adsorptions are carefully studied. To save the computational costs, limited numbers of Li_2S_4 and Li_2S_8 on borophenes are considered, as illustrated in Fig. 3(b). Obviously, the average binding energies of Li_2S_n on borophenes are much larger than those on electrolytes with the increase of sulfur loading, indicating that the Li_2S_n is hard to flee away from the borophenes surface even under high loading. Moreover, the average binding energies of Li_2S_n on borophenes have distinct difference with the different loading of sulfur, especially for the Li_2S_8 with high sulfur loading. Specifically, the average binding energies for one and four Li_2S_8

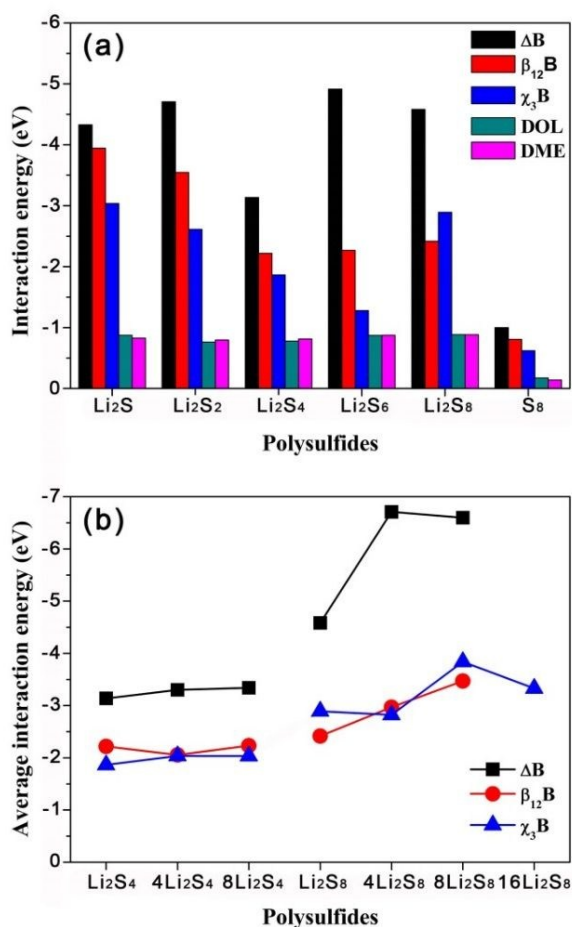


Fig. 3 (a) The interaction energies between Li_2S_n and borophenes, between Li_2S_n and solvent molecules. (b) The average interaction energies of multiple Li_2S_n on borophenes.

on ΔB are 4.58 eV and 6.71 eV, respectively, and the corresponding contents of sulfur are 16.5 wt% and 44.2 wt%. With the increase of sulfur loading in $8\text{Li}_2\text{S}_8$ on ΔB (sulfur loading is 61.3 wt%), the average binding energy is slightly decreased to 6.60 eV. Significantly, the sulfur loading is high up to ~80 wt% for $16\text{Li}_2\text{S}_8$ on $\gamma_3\text{B}$, where double-layered Li_2S_8 molecules are adsorbed on each side of borophene as in Fig. S5†. For $16\text{Li}_2\text{S}_8$ on $\gamma_3\text{B}$, the average adsorption energy of Li_2S_8 is 3.33 eV, with a much stronger interaction than the attraction from electrolyte, indicating that borophene should be an ideal option for practical applications of LSBs. It is worth noting that the sulfur loading can theoretically represent the initial utilization of sulfur, which should be slightly higher than experimental data for the neglected positive and negative factors, including lithium salts, additives, binders and other components in cells. Even so, our method in this work should be more scientific to qualitatively evaluate potential hosts, which has been used to predict the corresponding values on graphene (as described in the last part of ESI). The fully-optimized structures of multiple Li_2S_4 and Li_2S_8 on three borophene sheets are presented in Fig. S4†, where the Li_2S_n can be adsorbed strongly on the borophene surfaces resulting in some distorted structures on the surface. For the strong interaction of multiple Li_2S_n adsorption, we

calculated the charge density difference of $16\text{Li}_2\text{S}_8@x\text{B}$. As displayed in Fig. S6†, clearly lots of charges are accumulated between selected Li_2S_8 and other components of $16\text{Li}_2\text{S}_8@x\text{B}$, indicating strong Coulomb interactions among Li_2S_8 molecules. One should be noticed that, as reported² that sulfides rapidly dissolve in solvents, the interaction between Li_2S_8 molecules is not strong enough to avoid the dissolution. Therefore, the strong adsorption for multiple Li_2S_n should be attributed to the adsorption from borophene and the intermolecular interactions among adsorbed Li_2S_n .

Above calculations preliminarily demonstrate that Li_2S_n can be localized on borophenes. To visually display the competition of hosts and electrolytes, we simulated the dynamic behaviors of long-chain Li_2S_n between host materials and electrolytes (electrolytes@ Li_2S_n @borophene) by using FP-MD method. According to the experimental processes that sulfides are first mixed with host materials, long-chain Li_2S_n clusters are placed between borophenes and electrolytes in structural models (Fig. 1). The simulated structures at 300 K after 25 ps FP-MD simulations in Fig. S7† show that Li_2S_n ($n=4,8$) clusters are squished, and the same phenomena can be observed from the DFT-optimized Li_2S_n on borophene (Fig. S4†). Notably, Li_2S_n clusters won't tend to leave away from borophene surfaces or dissolve in electrolytes, agreeing well with the larger binding energies of Li_2S_n on borophene than that on electrolytes. In view of the results of DFT and FP-MD, the strong interaction between Li_2S_n and electrolyte are beneficial for the dissolution, while the introduced borophene sheets as host materials would greatly suppress such dissolution, even under high sulfur loading.

To deeply understand the interfacial competition between borophene and electrolytes, the interface-interactions between borophenes and Li_2S_n or electrolytes and Li_2S_n in electrolytes@ Li_2S_n @borophene are evaluated by further DFT calculations. Expectedly, the interface-interactions between borophenes and Li_2S_n are much higher than them between electrolytes and Li_2S_n . Specifically, the interface-interactions are 13.92 eV, 13.51 eV and 6.89 eV for ΔB , $\beta_{12}\text{B}$, and $\gamma_3\text{B}$ to Li_2S_8 , respectively, and the corresponding interface-interactions between electrolytes and Li_2S_8 are only 2.42 eV, 2.41 eV and 2.82 eV, respectively. Such strong interface-interactions can be attributed to the strong Coulomb interactions. In Fig. 4, the charge density differences of interfaces are displayed. Clearly, the charge accumulations are mainly distributed between S and B atoms, resulting in strong Coulomb interactions between borophene and Li_2S_8 . In contrast, little re-distributed charge can be observed between Li_2S_n and electrolyte at the selected isosurface level, implying a weaker Coulomb interaction between Li_2S_8 and electrolytes. In other words, the Coulomb interactions between Li_2S_n and electrolytes are limited, which agree well with relatively small binding energies between Li_2S_n and electrolytes. Overall, for borophene-based LSBs, it is the host materials rather than electrolytes dominantly affect the immobilization of sulfides on cathodes.

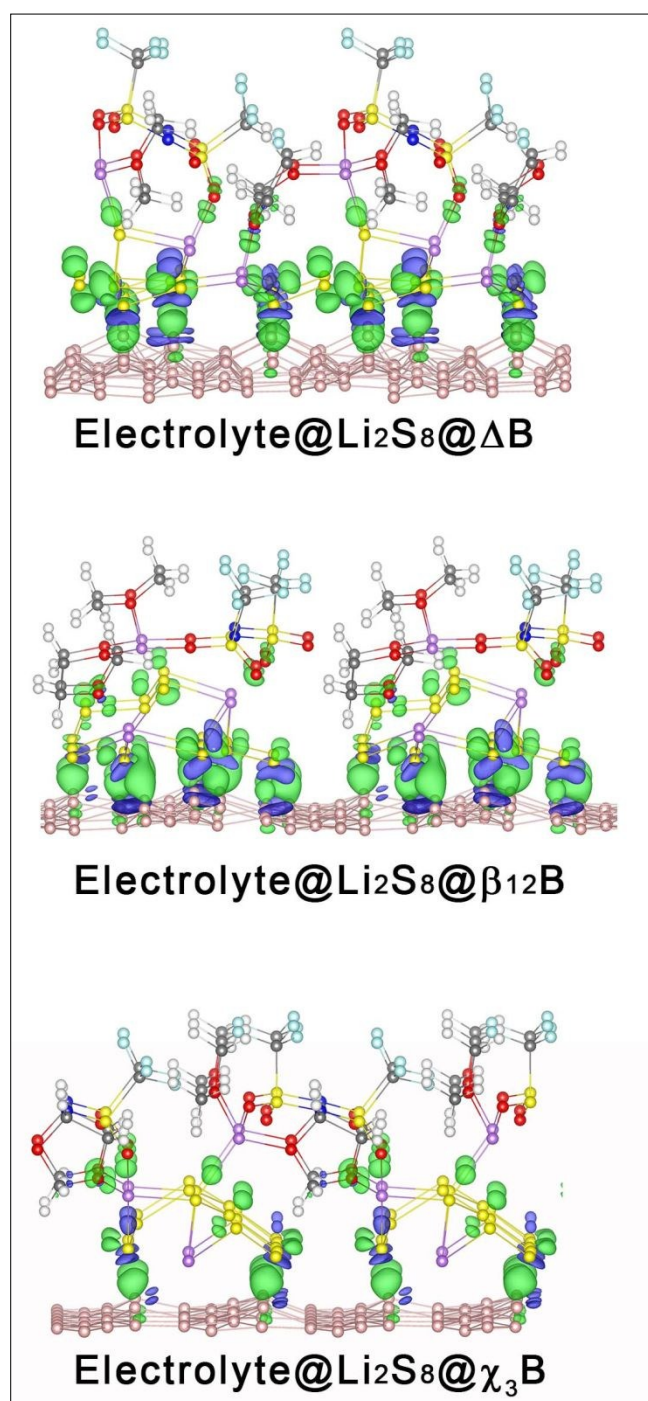


Fig. 4 The charge density difference of Li_2S_n on borophenes with electrolytes (DOL and DME) included. Green areas: charge accumulation, blue areas: charge depletion at the isosurface level of $0.025 \text{ e}/\text{\AA}^3$.

On the one hand, the geometric structures of grooves or pores in borophene sheets not only leads to the polarization of charge distribution as demonstrated to enhance the localization of Li_2S_n in cathodes,^{6, 35-37} but also provides more spaces for the accommodation of Li_2S_n , which can remedy the large volume expansion during electrochemical reaction process. On the other hand, the metallic borophene sheets with good conductivity are also satisfied candidates to be used in Li-S batteries. As a matter of fact, the conductivity of cathode can

be changed during charge/discharge cycles⁴⁷. Therefore, the conductivities of borophene-based cathodes should be evaluated, at least qualitatively. The distribution of density of state (DOS) in Fig. 5 shows that the cathodes based on $4\text{Li}_2\text{S}_n$ @borophenes ($n=4,8$) are all metallic with continuous electron densities at the Fermi level. Similar phenomenon also happens on that single Li_2S_n adsorbed on borophenes (Fig. S8[†]). Although, the conductivities of such cathodes get poorer and poorer with increasing the sulfur loading in Li_2S_n , for instance, there are no obvious band gaps for $8\text{Li}_2\text{S}_8$ @ ΔB and $8\text{Li}_2\text{S}_8$ @ $\chi_3\text{B}$ cathodes (Fig. S9[†]), and a band gap of 0.373 eV (Fig. S9[†]) for $8\text{Li}_2\text{S}_8$ @ $\beta_{12}\text{B}$, these narrow band gaps cannot greatly reduce the conductivity. Therefore, borophenes are the promising candidates for LSBs.

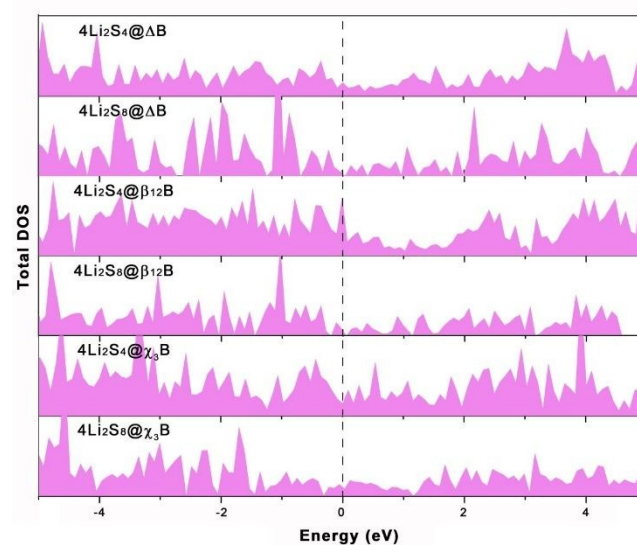


Fig. 5 The density of states of $4\text{Li}_2\text{S}_n$ ($n=4, 8$) on borophenes.

Computational Methods

Density functional theory (DFT) calculations were performed by the Vienna *ab initio* simulation package (VASP)⁴⁸ to optimize the geometric structures and obtain the electronic properties, which considered ion-electron interactions by the projected augmented wave method combined with exchange-correlation functional of Perdew-Burke-Ernzerhof in the generalized gradient approximation⁴⁹⁻⁵⁰. The criteria of 10^{-5} eV in energy, $10^{-2} \text{ eV}/\text{\AA}$ in force and 500 eV in cutoff energy were set in numerical calculations. For all 2×2 supercell models, the k -mesh grids were $5 \times 5 \times 1$ with a vacuum space of $20 \text{ \AA}$. To accurately describe the weak interactions of van der Waals, DFT-D3 dispersion correction method with Becke-Jonson damping⁵¹ was employed is more accurate for ground-state calculations.⁵² The first principles molecular dynamics (FP-MD) simulations were also carried out by the VASP code. At 300 K , FP-MD simulations lasted for 25 ps with a time step of 1 fs , and the Nosé algorithm was taken to control the temperature. The binding energy (E_b) of Li_2S_n is calculated by the equation, $E_b = E_{\text{total}} - E_{\text{borophene}} - E_{\text{LS}}$, in which E_{total} , $E_{\text{borophene}}$ and E_{LS} respectively represent the total energy of Li_2S_n adsorbed

borophene, the energies of borophene and Li_2S_n . The interface-interaction (E_{ij}) is calculated by the equation, $E_{ij} = E_{AB} - E_{AB-B} - E_{AB-A}$, in which the E_{AB} , E_{AB-B} and E_{AB-A} respectively represent the total energy of AB interface system, total energy of A in AB system, and total energy of B in AB system. The charge density difference (ρ_d), i.e., the charge redistribution caused by the adsorption of Li_2S_n clusters, were calculated by the equation, $\rho_d = \rho_{\text{total}} - \rho_{\text{borophene}} - \rho_{\text{LS}}$, where ρ_{total} , $\rho_{\text{borophene}}$ and ρ_{LS} respectively represent the total charge density of Li_2S_n adsorbed borophene, the charge densities of borophene and Li_2S_n .

To consider the utilization of sulfur, we calculated the sulfur loading from Li_2S_8 as following:

$$\text{Mass (wt\%)} = \frac{m_s}{m_s + m_b} \times 100\%$$

where m_s represents the total mass of sulfur atoms in adsorbed Li_2S_n molecules, and m_b represents the mass of borophene. The sulfur loading from this formula is for the theoretical initial utilized sulfur.

Conclusions

In summary, from the computational point of view, we demonstrate an interfacial competition between borophene-based cathode and electrolyte in LSBs, and the conqueror is borophene sheet as host material that dominates the multiple-sulfide immobilization. Our calculated results suggest that the polarized surfaces of three borophenes provide appropriate sites to anchor sulfides. The strong binding of Li_2S_n on borophenes can prevent the dissolution of polysulfides. These borophenes as host materials can confine under high sulfur loading around 80 wt%, which can satisfy the relevant requirements for practical applications. The good conductivities of borophenes can overcome the conductivity problems from sulfur and its compounds. For borophene-based or B-doped host materials, S atoms are favourably adsorbed by B atoms, which is very similar to that S atoms are firstly bonded by O or metal atoms in metal oxides,⁵³⁻⁵⁴ resulting in a low polysulfide dissolution. Also, keep in mind that the B atom is fascinating for high energy density for its light mass, so we expect that B-rich 2D materials truly deserve much attention in future experimental and computational studies.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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